

DEVOPS MASTER COURSE

Volume 7B

Azure Core Infrastructure

Chapters 11-20

VM SCALE SETS | HUB-SPOKE | FIREWALL | APPLICATION DELIVERY | PRIVATE LINK | STORAGE |
HYBRID CONNECTIVITY | BACKUP | DISASTER RECOVERY

Production-grade Azure compute, networking, storage, hybrid connectivity, recovery,
troubleshooting, labs, and interviews

2026 Edition

About This Part

Volume 7B develops the infrastructure layer that supports production workloads in Microsoft Azure. Part A established identity, governance, virtual-machine, and virtual-network fundamentals. Part B turns those primitives into scalable compute fleets, segmented network topologies, secure application-delivery paths, resilient storage systems, hybrid connectivity, backup, and disaster-recovery capabilities.

The engineering objective is not simply to create resources. It is to define failure domains, traffic paths, recovery boundaries, ownership, health signals, upgrade methods, and evidence before a workload is exposed to users. Every chapter connects design intent to implementation, validation, troubleshooting, security, and cleanup.

Safety and Cost Controls

- Use a sandbox subscription or an organization-approved training subscription. Verify regional availability, quotas, pricing, and feature status before deployment.
- Create budgets and expiry tags before provisioning scale sets, gateways, firewalls, Front Door profiles, Application Gateways, VPN gateways, ExpressRoute resources, or recovery services.
- Use managed identities and Azure RBAC. Never place credentials, storage keys, shared access signatures, or private keys in scripts, screenshots, source control, or VM custom data.
- Prefer Azure Bastion, private connectivity, and least-privilege network rules. Do not expose RDP or SSH to the entire internet.
- Run destructive operations only in approved training environments. Record cleanup evidence because disks, snapshots, public IP addresses, gateways, backups, and replication can continue to generate charges after compute is stopped.

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VOLUME 7B | CHAPTER 11

Chapter 11 - Advanced Azure Virtual Machines and Fleet Operations

Standardizing VM lifecycle, images, availability, patching, access, and operational evidence

Learning Objectives

- Design virtual machines for availability-zone, availability-set, and maintenance requirements.
- Standardize images, extensions, cloud-init, custom script execution, and Azure Compute Gallery usage.
- Operate VMs through managed identities, Azure Bastion, Update Manager, Run Command, and monitoring agents.
- Diagnose provisioning, boot, extension, disk, quota, and connectivity failures without weakening security.

11.1 Production VM Design Starts with the Workload

A production VM is a managed operating-system boundary, not a permanent hand-built server. The architecture must identify the workload process, expected traffic, data persistence, recovery point objective, recovery time objective, maintenance window, administrative path, and replacement method. A VM that cannot be recreated from an image plus configuration is a hidden single point of operational knowledge.

Choose a VM family from CPU, memory, storage throughput, network bandwidth, accelerator, and licensing requirements. Size testing must use application-level measurements rather than CPU percentage alone. A low CPU value can hide blocked storage, connection limits, memory pressure, or serialization bottlenecks.

Decision	Use when	Critical design check
Availability zones	The region supports zones and the workload requires datacenter-failure resilience.	Deploy multiple instances across zones and validate zonal dependencies.
Availability set	A nonzonal regional workload requires separation across fault and update domains.	Confirm the workload can tolerate shared regional failure.
Single VM	Development, disposable tooling, or explicitly accepted lower availability.	Document the risk and protect data independently.
Dedicated host	Hardware isolation, licensing, or compliance requires dedicated physical hosts.	Plan host groups, fault domains, capacity, and cost.
Spot VM	Interruptible batch, CI workers, rendering, or resilient stateless tasks.	Handle eviction and persist checkpoints outside the VM.

11.2 Images, Configuration, and Drift

Azure Compute Gallery provides versioned image distribution and replication. A mature image pipeline builds from source, applies updates, runs vulnerability and functional tests, signs or records provenance, publishes an immutable version, and promotes that version through environments. Configuration that changes frequently should be applied after boot by a controlled configuration mechanism rather than baked into every image.

```
az vm image list --publisher Canonical --offer ubuntu-24_04-lts --all -o table
az vm show -g rg-core -n vm-app-01 --query
'{id:id,size:hardwareProfile.vmSize,identity:identity.type}'
```

11.3 Secure Administration and Fleet Operations

- Use Azure Bastion, private IP administration, VPN/ExpressRoute, or just-in-time access instead of broad public management ports.
- Use system-assigned or user-assigned managed identities for Azure resource access. Avoid credentials in files or environment variables.
- Use Azure Update Manager to assess and orchestrate guest patching according to workload maintenance policy.
- Use Run Command and serial console as controlled break-glass mechanisms with activity logging and time-bounded privilege.
- Collect Azure platform metrics, guest telemetry, boot diagnostics, activity logs, and application health signals before an incident occurs.

11.4 Operational Failure Domains

Failure	Evidence	Engineering response
ProvisioningState failed	Deployment operations, activity log, extension status	Correct invalid parameters, quota, image, identity, or networking dependency; redeploy from source.
VM is running but unreachable	NIC effective rules/routes, NSG flow, guest firewall, Bastion diagnostics	Trace from DNS through route, NSG, load balancer, and guest service.
Extension timeout	Instance view, extension logs, package repositories, DNS and proxy path	Make scripts idempotent, noninteractive, time bounded, and externally observable.
Boot failure	Boot diagnostics screenshot, serial log, disk state	Repair through recovery VM or restore; do not disable diagnostic controls.
Quota or capacity failure	Usage quotas, SKU availability, zonal capacity	Request quota early and provide alternate sizes/zones in deployment design.

Enterprise pattern Treat VMs as replaceable members of a controlled fleet. Persist business data outside the OS disk, version every image and configuration input, and define rollback before deployment.

Hands-on Lab 11: Build a Secure, Reproducible Azure VM

Objective Deploy a Linux VM without broad internet administration, attach a managed identity, collect diagnostics, and prove that the workload can be recreated.

Prerequisites

- Azure CLI in Cloud Shell
- Contributor on a sandbox resource group and permission to assign a managed identity if required
- An approved region and a cost budget

Implementation

1. Create a tagged resource group and virtual network with a private workload subnet.
2. Create a network security group that permits only application traffic required by the lab; do not permit 0.0.0.0/0 SSH.
3. Deploy a Linux VM with a system-assigned managed identity, boot diagnostics, and SSH key authentication.
4. Use Bastion or another approved private administrative path to connect.
5. Install a small web service through cloud-init or a repeatable script and store the script in source control.
6. Use Azure CLI to capture VM identity, size, zone, NIC, disk, extension, and power-state evidence.
7. Delete and recreate the VM from the same documented inputs; confirm the service returns.

Validation Evidence

- The VM has no broadly exposed management port.
- Managed identity is enabled and no secrets are embedded in custom data.
- Boot diagnostics and activity-log evidence are captured.
- The recreated VM produces the same validated result.

Cleanup

- Remove VM, NIC, public IP if used, disks, Bastion if created only for the lab, and the resource group.
- Confirm no orphaned managed disks or snapshots remain.

Stretch challenge Publish a versioned custom image to Azure Compute Gallery and deploy the VM from that image.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Deployment fails with allocation error	Requested SKU or zone lacks capacity	Review deployment error, SKU availability, and quotas	Retry an approved alternate zone or size; design fallback options.
Managed identity receives 403	Role missing, wrong scope, or propagation delay	Inspect role assignment and token audience	Assign the narrow role at the correct scope and retest after propagation.
Custom script repeats harmful actions	Script is not idempotent	Review extension status and script logs	Add state checks, safe retries, explicit exit codes, and versioned artifacts.

Interview Questions and Model Answers

1. Why should production VMs be replaceable?

Replacement removes configuration drift, supports deterministic recovery, and allows tested image and configuration pipelines.

2. When are availability zones preferable to availability sets?

Zones protect against datacenter-level failure within a region, while availability sets separate instances across platform fault and update domains without zonal isolation.

3. Why avoid secrets in custom data?

Custom data can be retained or exposed to privileged users and tooling; use managed identity and a secret store instead.

4. What evidence is required before patching a VM fleet?

Asset scope, health baseline, maintenance window, backup or rollback path, staged deployment result, patch compliance, and post-change application validation.

5. How do you troubleshoot a running but unreachable VM?

Validate name resolution, routes, effective NSG rules, load-balancer path, NIC state, guest firewall, listening service, and application logs in that order.

Chapter Summary

- VM reliability depends on replacement, not manual repair alone.
- Images and configuration require versioning, testing, promotion, and rollback.
- Secure administration uses private paths, managed identity, and logged break-glass operations.
- Diagnostics must be deployed before failure.

VOLUME 7B | CHAPTER 12

Chapter 12 - Virtual Machine Scale Sets and Autoscale

Building elastic, health-aware, upgradeable compute fleets

Learning Objectives

- Compare Flexible and Uniform orchestration and select an appropriate fleet model.
- Configure health probes, application health, autoscale rules, scale-in policy, and instance repair.
- Design rolling upgrades and image promotion with capacity and rollback protection.
- Troubleshoot autoscale, unhealthy-instance, load-balancer, and capacity behavior.

12.1 Scale Sets Are an Orchestration Boundary

A Virtual Machine Scale Set defines a fleet and the rules by which instances are created, distributed, monitored, scaled, upgraded, and removed. It is most valuable when application instances are replaceable and business state is externalized. The fleet model must include health semantics: infrastructure power state is insufficient because a VM can be running while the application is unavailable.

Capability	Flexible orchestration	Uniform orchestration
Instance model	Supports a broader range of VM patterns and individual VM management.	Instances are managed from a common scale-set model.
Use case	Modern VM fleets requiring flexibility, zonal distribution, and mixed management.	Highly uniform large fleets and established uniform workflows.
Health integration	Application Health extension or load-balancer health is central.	Health probes and extension integration support upgrades and repair.
Operational principle	Recommended starting point for many new VM deployments; verify feature needs.	Use when specific uniform capabilities or existing design require it.

12.2 Autoscale Is a Feedback-Control System

Autoscale uses measurements to change capacity. Every policy therefore needs a valid signal, an observation window, threshold, action, cooldown, minimum, maximum, and default capacity. Scale-out should protect latency before saturation. Scale-in should be slower and should account for connection draining, queue depth, state transfer, and minimum fault-domain coverage.

Signal	Strength	Risk if used alone
CPU percentage	Simple and widely available.	Does not reveal queue backlog, storage latency, or external dependency saturation.
Request rate	Tracks demand for request-driven services.	Different request types may have different cost.
Queue depth / messages per instance	Strong for worker fleets.	Requires correct age and poison-message handling.

Signal	Strength	Risk if used alone
Application latency or custom metric	Closest to user impact.	Metric pipeline delay and cardinality must be controlled.
Schedule	Useful for known business cycles.	Cannot react to unexpected demand by itself.

12.3 Rolling Upgrades and Health Protection

A rolling upgrade changes instances in controlled batches. Safe operation requires a trustworthy health signal, sufficient spare capacity, tested startup time, maximum unhealthy limits, pause time, and rollback decision. MaxSurge can create temporary capacity during upgrades, which improves availability but affects quotas and cost. Scale-in policy must preserve required zone distribution and handle instance protection for critical work.

```
az vmss list -g rg-core -o table
az monitor autoscale list -g rg-core -o table
az vmss get-instance-view -g rg-core -n vmss-web
```

Production gate Do not enable automatic image upgrades until the application health signal reliably distinguishes a healthy service from a merely running VM.

Hands-on Lab 12: Deploy a Health-Aware VM Scale Set

Objective Create a zone-aware web fleet behind a load balancer, configure autoscale, and perform a controlled image or configuration update.

Prerequisites

- Completed Chapter 11 concepts
- Quota for at least three small VM instances
- A test script that generates load

Implementation

8. Create a scale set using an approved orchestration mode and distribute instances across supported zones.
9. Install a web service that exposes a dedicated health endpoint.
10. Attach the scale set to a Standard Load Balancer or Application Gateway backend.
11. Configure application health monitoring and confirm unhealthy behavior is detected.
12. Create autoscale settings with minimum, default, maximum, scale-out, and conservative scale-in rules.
13. Generate controlled load and record capacity and metric changes.
14. Apply a versioned configuration update using a rolling strategy and capture prechange, during-change, and postchange health.

Validation Evidence

- Traffic remains available during a healthy rolling update.
- Autoscale responds to the intended metric and respects limits.
- An intentionally unhealthy instance is excluded or repaired according to design.
- Scale-in does not remove protected or required instances unexpectedly.

Cleanup

- Remove autoscale settings, scale set, load balancer components, public IP, and resource group.
- Confirm managed disks and network interfaces were deleted.

Stretch challenge Use a custom application metric and compare its behavior with CPU-based autoscale.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Autoscale does not trigger	Metric dimension, threshold, time grain, or permissions are wrong	Inspect metric chart and autoscale run history	Correct the rule and validate with sustained test load.
Rolling upgrade stalls	Health probe or extension reports unhealthy	Inspect instance health and application startup logs	Fix health semantics; do not bypass the health gate.
New instances fail bootstrap	Image, package repository, DNS, or identity dependency fails	Review extension and cloud-init logs	Use versioned artifacts, private repositories, retries, and dependency health checks.

Interview Questions and Model Answers

1. Why is CPU-only autoscale often inadequate?

CPU can remain low while queues, latency, memory, storage, or downstream dependencies become saturated.

2. What makes a health endpoint trustworthy?

It checks the service path needed to accept work without depending on optional components that create false failures.

3. Why should scale-in be conservative?

Removal can terminate active work and reduce zonal resilience; scale-in must allow draining and preserve minimum capacity.

4. What is the purpose of MaxSurge?

It adds temporary capacity during rolling upgrades so old instances need not be removed before replacements are healthy.

5. How would you prove autoscale works?

Use controlled load, record the metric and evaluation windows, verify instance count changes, validate traffic, and confirm cooldown and scale-in behavior.

Chapter Summary

- Scale sets define fleet lifecycle and health, not just instance count.
- Autoscale is a measured feedback loop that needs workload-specific signals.
- Rolling upgrades require valid health, spare capacity, staged promotion, and rollback.
- Fleet behavior must be tested under failure and scale-in conditions.

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Chapter 13 - Enterprise Virtual Networking and Hub-Spoke Architecture

Designing address spaces, subscriptions, peering, routing, and shared network services

Learning Objectives

- Design nonoverlapping address plans and subnet boundaries for enterprise Azure estates.
- Compare customer-managed hub-spoke, Azure Virtual WAN, full mesh, and isolated workload patterns.
- Explain VNet peering, gateway transit, route propagation, DNS, and shared-service placement.
- Use Network Watcher and effective-route evidence to troubleshoot connectivity.

13.1 Address Space Is a Long-Lived Architecture Decision

IP address allocation survives individual applications. Overlapping networks block simple peering and complicate hybrid connectivity, acquisitions, and multi-region growth. Enterprise planning reserves ranges for regions, environments, hubs, spokes, platform services, gateways, and future expansion. Subnets should represent operational or security boundaries rather than every application component by default.

Subnet category	Typical contents	Design requirement
GatewaySubnet	VPN or ExpressRoute gateway	Dedicated name and sufficient headroom; do not deploy unrelated resources.
AzureFirewallSubnet	Azure Firewall	Dedicated subnet sized according to service guidance.
AzureBastionSubnet	Azure Bastion	Dedicated subnet and required service traffic.
Application Gateway subnet	Application Gateway instances	Dedicated subnet with scaling headroom and compatible controls.
Workload subnet	VMs, scale sets, private endpoints as designed	NSG, routes, DNS, service delegation, and IP capacity documented.
Private endpoint subnet	Private endpoint NICs where separation is desired	DNS ownership and network policy behavior defined.

13.2 Hub-Spoke and Virtual WAN

In a customer-managed hub-spoke topology, one or more hub VNets host shared connectivity and security services while spoke VNets isolate workloads. VNet peering is low latency but is nontransitive; a spoke does not automatically reach another spoke through the hub. Transit requires routing through a firewall, network virtual appliance, gateway, or managed Virtual WAN hub according to the chosen architecture.

Topology	Strength	Trade-off
Customer-managed hub-spoke	Explicit control over hub services, routing, and appliances.	More route, peering, lifecycle, and scale management.

Topology	Strength	Trade-off
Azure Virtual WAN	Managed hubs and large-scale branch, user, ExpressRoute, and VNet connectivity.	Different routing and cost model; requires deliberate migration and policy design.
Full mesh peering	Simple for a small number of VNets.	Connection count and policy complexity grow quickly.
Isolated VNet	Strong separation for special workloads.	Shared services and operations need explicit private paths.

13.3 Routing and Gateway Transit

- System routes provide default VNet, peering, and internet behavior. User-defined routes override selected paths.
- Gateway transit lets a spoke use a hub gateway when peering settings and topology support the design.
- BGP-learned routes enter through VPN Gateway, ExpressRoute, Route Server, or Virtual WAN scenarios and require route ownership discipline.
- Peering options such as forwarded traffic and remote gateways change transit behavior; document them as architecture decisions.
- Use effective routes and next-hop diagnostics on a NIC to validate the actual data plane rather than relying on diagrams alone.

```
az network vnet peering list -g rg-network --vnet-name vnet-spoke-app -o table
az network nic show-effective-route-table -g rg-app -n nic-app-01 -o table
az network watcher show-next-hop -g rg-network --vm vm-app-01 --source-ip 10.20.1.4 --dest-ip 10.10.2.4
```

Hands-on Lab 13: Build and Validate a Hub-Spoke Topology

Objective Create a hub and two spokes, configure peering and controlled transit, and prove the intended routes.

Prerequisites

- Permission to create VNets, peerings, NSGs, route tables, and small test VMs
- A documented nonoverlapping CIDR plan

Implementation

15. Create one hub VNet and two spoke VNets with dedicated, nonoverlapping address ranges.
16. Create required platform and workload subnets with appropriate naming and tags.
17. Peer each spoke to the hub with settings documented before implementation.
18. Deploy small test VMs without broad public access and use Bastion or Run Command for tests.
19. Validate hub-to-spoke connectivity and demonstrate that spoke-to-spoke is not automatically transitive.
20. Add an approved transit path using a firewall/NVA route design or document a simulation if cost prevents deployment.
21. Capture effective routes, effective security rules, DNS results, and Network Watcher connection evidence.

Validation Evidence

- No CIDR ranges overlap.

- Peerings show connected state.
- Observed traffic matches the documented transit design.
- Effective routes identify the correct next hop for every tested destination.

Cleanup

- Delete test VMs, peerings, routes, NSGs, VNets, and resource groups.
- Confirm diagnostic resources created for the lab are removed if no longer needed.

Stretch challenge Extend the design to a second region and write the routing and failover decision record.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Peering is connected but traffic fails	NSG, UDR, guest firewall, or forwarded-traffic setting blocks path	Inspect effective routes and rules on both NICs	Correct the exact control; avoid broad allow rules.
Spoke cannot use hub gateway	Gateway transit and remote gateway settings are inconsistent	Inspect both peering directions and gateway state	Set complementary options and validate supported topology.
New VNet cannot peer	Address spaces overlap	Compare all connected prefixes including on-premises	Re-number before production or use an approved translation strategy.

Interview Questions and Model Answers

1. Why is VNet peering called nontransitive?

Connectivity between A and B and between B and C does not automatically create connectivity between A and C.

2. Why centralize selected network services in a hub?

It reduces duplicate gateways and security appliances and creates shared inspection, DNS, connectivity, and operations boundaries.

3. When can Virtual WAN be preferable?

When the organization needs Microsoft-managed hub infrastructure and scalable branch, remote-user, ExpressRoute, VPN, and VNet transit.

4. What is the first artifact in enterprise network design?

A governed address plan that accounts for regions, environments, hybrid networks, services, and growth.

5. How do you prove an Azure route?

Inspect effective routes and use next-hop or connection troubleshooting from the actual source interface.

Chapter Summary

- Address planning and route ownership are foundational enterprise controls.
- Hub-spoke centralizes connectivity while spokes isolate workloads.
- Peering is nontransitive and must not be mistaken for automatic transit.
- Effective data-plane evidence is more authoritative than a diagram.

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Chapter 14 - Network Security, Routing, Firewall, and Secure Administration

Layering NSGs, ASGs, UDRs, firewalls, DDoS controls, Bastion, and diagnostics

Learning Objectives

- Explain stateful NSG processing, priorities, service tags, and application security groups.
- Design centralized ingress and egress inspection with UDRs and Azure Firewall.
- Use Bastion, just-in-time access, DDoS protection, and private administration patterns.
- Troubleshoot denied traffic using effective rules, flow evidence, next hop, DNS, and application state.

14.1 Layer Controls by Function

Control	Primary function	Common mistake
Network security group	Stateful L3/L4 filtering at subnet or NIC.	Using broad IP rules instead of workload roles and service tags.
Application security group	Groups NICs by application role for NSG rules.	Assuming it operates across unrelated VNets without design validation.
Route table / UDR	Selects next hop for traffic.	Forcing traffic to an appliance without return-path symmetry.
Azure Firewall	Central stateful network and application filtering with managed scale.	Deploying it without route, DNS, logging, and policy ownership.
Web Application Firewall	Protects HTTP(S) applications from web attacks.	Treating it as a replacement for secure application code and patching.
DDoS protection	Helps protect public IP resources from volumetric attacks.	Assuming it replaces application-layer rate limiting or WAF.

14.2 NSG and ASG Design

NSG rules are evaluated by priority and use five-tuple fields plus service tags or ASGs. Because NSGs are stateful, return traffic for an allowed connection is automatically permitted. Use subnet-level policies for broad tier boundaries and application security groups to express roles such as web, application, and data instead of maintaining fragile individual IP lists.

A deny result is not solved by adding an unrestricted allow. Determine whether the packet reached the expected NIC, which route was selected, which NSGs apply, whether the service is listening, and whether the return path is symmetric. Effective security rules provide the merged result for a NIC.

14.3 Central Inspection and Egress Governance

A common pattern places Azure Firewall in a hub and associates spoke subnets with route tables that send selected traffic to the firewall. Firewall Policy separates rule collections and can be managed across environments. DNS proxy, threat intelligence, application rules, network rules, and logging must be designed

together. Forced tunneling to on-premises is a compliance choice that introduces dependency and latency, so failure behavior must be tested.

```
az network nic list-effective-nsg -g rg-app -n nic-app-01 -o json
az network nic show-effective-route-table -g rg-app -n nic-app-01 -o table
az network watcher test-ip-flow -g rg-network --vm vm-app-01 --direction Outbound --protocol
TCP --local 10.20.1.4:60000 --remote 10.30.1.4:443
```

Zero-trust interpretation Network location is not identity. Use network segmentation to reduce attack paths, but also require strong workload identity, authorization, encryption, secret management, patching, and telemetry.

Hands-on Lab 14: Implement Tiered Network Security

Objective Create web, application, and data roles, enforce least-privilege paths, and validate denied flows without using broad exceptions.

Prerequisites

- A sandbox VNet with three workload subnets or a smaller simulated equivalent
- Network Contributor and VM access through an approved method

Implementation

22. Create application security groups for web, application, and data roles.
23. Associate test NICs with the appropriate ASGs.
24. Create subnet NSGs that allow only web-to-application and application-to-data ports required by the lab.
25. Deny direct web-to-data access and all broad management access.
26. Add a route table for controlled egress through a lab firewall/NVA, or document the exact production route if cost prevents deployment.
27. Use Bastion or Run Command to test allowed and denied connections.
28. Capture effective NSGs, effective routes, test-ip-flow, service listener, and firewall evidence.

Validation Evidence

- Required tier flows succeed.
- Prohibited lateral and direct data-tier flows fail.
- No rule allows unrestricted SSH/RDP from the internet.
- Every allow rule has a documented owner and purpose.

Cleanup

- Delete test VMs, NICs, ASGs, NSGs, routes, firewall/NVA resources, and the resource group.
- Verify public IPs and firewall policies do not remain.

Stretch challenge Create an Azure Policy concept that audits subnets without NSGs and public IPs without approved tags.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Allowed rule exists but traffic fails	Higher-priority deny, route, guest firewall, or service state	Inspect effective rules, next hop, and listener	Correct the first failing layer rather than adding broad access.
Traffic reaches appliance but response fails	Asymmetric return route or SNAT behavior	Capture route and firewall logs in both directions	Fix route symmetry and appliance forwarding/SNAT configuration.
Private VM administration fails	Bastion subnet, NSG, route, or target guest port is incorrect	Validate Bastion requirements and target effective rules	Restore required service traffic and least-privilege guest access.

Interview Questions and Model Answers

1. Are NSGs stateful?

Yes. Return traffic for an allowed connection is automatically permitted, though the initial flow must satisfy effective rules.

2. Why use ASGs?

They let NSG rules reference application roles rather than changing IP addresses, improving intent and maintainability.

3. What is the purpose of a UDR to a firewall?

It overrides system routing so selected traffic traverses the inspection next hop.

4. Why can forced tunneling create risk?

Azure workload internet access becomes dependent on the on-premises path and inspection capacity, creating latency and failure coupling.

5. What is the safest response to an unexpected deny?

Collect effective rules, routes, DNS, and service evidence; correct the narrow cause instead of adding a broad allow.

Chapter Summary

- NSGs, ASGs, routes, firewalls, and WAF solve different control problems.
- Central inspection requires symmetric routing, DNS, policy, logs, and capacity.
- Secure administration avoids broad public management ports.
- Troubleshooting follows the complete packet path and application state.

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Chapter 15 - Azure Load Balancing and Global Application Delivery

Selecting Layer 4, regional Layer 7, global HTTP acceleration, and DNS routing

Learning Objectives

- Compare Azure Load Balancer, Application Gateway, Front Door, and Traffic Manager.
- Design probes, backend pools, listener rules, TLS termination, WAF, and origin security.
- Build regional and multi-region application-delivery paths with failure testing.
- Diagnose unhealthy backends, routing errors, certificates, DNS, SNAT, and asymmetric paths.

15.1 Start with Protocol, Scope, and Processing Layer

Service	Layer and scope	Best fit
Azure Load Balancer	Regional Layer 4 TCP/UDP	High-performance network load balancing for VMs and scale sets.
Application Gateway	Regional Layer 7 HTTP(S)	Host/path routing, TLS termination, autoscaling, and regional WAF.
Azure Front Door	Global Layer 7 HTTP(S) edge service	Global acceleration, anycast entry, origin failover, caching, and WAF.
Traffic Manager	Global DNS-based traffic distribution	Protocol-independent endpoint selection where clients connect directly to endpoints.

A service name containing load balancing does not make the products interchangeable. The decision depends on protocol, public or private exposure, regional or global scope, proxy versus DNS behavior, TLS ownership, web application firewall requirements, session behavior, source IP needs, and failover objectives.

15.2 Health Probes and Backend Readiness

Health probes decide whether a backend receives new traffic. A probe should be inexpensive, deterministic, and representative of the service ability to accept work. It should not report healthy before the application is ready, and it should not depend on optional downstream systems that create cascading false failures. Backend connection draining and graceful shutdown are required during scale-in and deployment.

15.3 TLS, WAF, and Origin Protection

- Terminate TLS at the layer that owns routing and certificate lifecycle; re-encrypt to origins when required.
- Use managed certificates or Key Vault integration where supported, and monitor expiration and domain validation.
- Enable WAF in detection mode first, analyze false positives, then promote governed rules to prevention.
- Restrict origins so clients cannot bypass Front Door or Application Gateway controls.
- Capture access logs, health logs, WAF events, backend latency, response codes, and dependency telemetry.

15.4 Multi-Region Patterns

A global HTTP workload commonly uses Front Door to select healthy regional origins, with Application Gateway or a regional application platform behind each origin. Non-HTTP workloads may use Traffic Manager for DNS

routing and regional Load Balancers. DNS-based failover is affected by caching and client behavior, while proxy-based global services retain control of the connection path. Recovery design must test state, data replication, and dependency readiness, not only endpoint health.

```
az network lb list -g rg-edge -o table
az network application-gateway show-backend-health -g rg-edge -n agw-app
az afd endpoint list -g rg-edge --profile-name afd-global -o table
```

Hands-on Lab 15: Build a Regional Application Delivery Path

Objective Publish a two-instance application through an appropriate Azure load-balancing service and prove health-based routing and controlled failure.

Prerequisites

- Two backend instances or scale-set instances
- A DNS name or test host header if using Layer 7
- Budget awareness for Application Gateway or Front Door

Implementation

29. Deploy two identical backend services with instance identity shown in the response.
30. Create a Standard Load Balancer for TCP or Application Gateway for HTTP(S), based on the selected design.
31. Configure backend pool, frontend, health probe, and rule/listener.
32. Restrict backend access so traffic follows the intended delivery path.
33. Validate distribution and record health status.
34. Stop the application on one backend and measure detection and traffic continuity.
35. Restore the backend and verify reintegration; document TLS and WAF production enhancements.

Validation Evidence

- Only healthy backends receive new traffic.
- Backend identity proves traffic distribution.
- Direct backend access is restricted according to design.
- Health and access logs support the observed behavior.

Cleanup

- Delete delivery service, public IPs, backends, certificates created only for training, and resource group.
- Confirm DNS records no longer reference deleted endpoints.

Stretch challenge Add a second region and design Front Door health, priority, and origin security without creating unnecessary cost.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Backend is unhealthy	Probe path/port/host, NSG, route, or service readiness is wrong	Inspect backend health and test probe locally	Fix probe semantics and exact network path.

Symptom	Likely cause	Investigation	Corrective action
502 or gateway error	Origin connection, TLS, DNS, timeout, or backend response fails	Review gateway/Front Door diagnostics and origin logs	Correct hostname, certificate trust, response time, or origin health.
Clients bypass WAF	Origin remains publicly reachable without restrictions	Test direct origin endpoint and review NSG/firewall rules	Allow only trusted delivery-service paths or private origin connectivity.

Interview Questions and Model Answers

1. When would you choose Load Balancer over Application Gateway?

For regional TCP/UDP Layer 4 distribution where HTTP-aware routing and WAF are not required.

2. How does Traffic Manager differ from Front Door?

Traffic Manager returns DNS answers and clients connect directly; Front Door is a global HTTP(S) reverse proxy and edge service.

3. What makes a bad health probe?

A probe that is expensive, reports healthy before readiness, or fails because an optional dependency is unavailable.

4. Why protect origins?

Otherwise attackers or users can bypass WAF, routing, TLS, and rate controls at the approved entry point.

5. How do you validate failover?

Induce a controlled backend or regional failure, observe health detection and user behavior, and validate state and data dependencies.

Chapter Summary

- Load-balancing selection begins with protocol, scope, and processing layer.
- Health semantics determine safe routing, deployment, and failover.
- WAF and TLS controls must be paired with origin restrictions.
- Multi-region routing is useful only when state and dependencies can recover.

VOLUME 7B | CHAPTER 16

Chapter 16 - Azure DNS, Private DNS, Private Link, and Service Access

Controlling names and private paths for platform services

Learning Objectives

- Operate public DNS zones, records, aliases, delegation, TTL, and DNSSEC considerations.
- Design Azure Private DNS zones, VNet links, hybrid resolution, and Private Resolver.
- Compare private endpoints, service endpoints, public network restrictions, and Private Link Service.
- Diagnose split-horizon, stale records, private endpoint, proxy, and hybrid DNS failures.

16.1 DNS Is Part of the Application Control Plane

DNS determines which endpoint clients attempt to reach. A correct IP route is useless when names resolve incorrectly. Public zones require controlled delegation, change approval, record ownership, TTL management, certificate coordination, and monitoring. Private zones require VNet links and hybrid resolution architecture. Treat DNS changes as production releases with rollback.

Record or feature	Purpose	Operational note
A / AAAA	Maps a name to IPv4 / IPv6 address.	Avoid manual records for ephemeral endpoints when alias integration is available.
CNAME	Maps one name to another canonical name.	Cannot generally coexist with other data at the same name and is unsuitable at zone apex in traditional DNS.
Alias record	Azure-aware pointer to supported resource.	Tracks selected Azure resource lifecycle without exposing a fixed IP.
TXT	Verification, policy, and service metadata.	Protect ownership because records can affect domain trust.
Private DNS zone	Resolves private names for linked VNets.	VNet links and auto-registration behavior require explicit ownership.
Private Resolver	Managed inbound/outbound DNS forwarding.	Useful for hybrid and multi-VNet resolution without self-managed DNS VMs.

16.2 Private Endpoints and DNS Integration

A private endpoint is a network interface in a VNet with a private IP that maps to a supported service through Azure Private Link. The network path remains on the Microsoft backbone, but successful private access depends on DNS returning the private endpoint IP. Each service uses documented private DNS zone names. Public network access should be restricted only after private connectivity and all clients are validated.

Method	Traffic path and identity	Use case
Public endpoint with firewall	Public service endpoint; access restricted by network rules and identity.	Approved internet or Microsoft network access with controlled sources.
Service endpoint	Extends VNet identity to selected service public endpoints.	Simple subnet-scoped service access where private IP endpoint is not required.
Private endpoint	Private IP in the consumer VNet maps to service.	Private service access, public exposure reduction, and granular approval.
Private Link Service	Publishes a customer service privately to consumers.	Provider-owned services exposed privately across VNets or tenants.

16.3 Hybrid DNS Resolution

On-premises clients accessing Azure private endpoints need conditional forwarding toward Azure. Azure clients resolving on-premises names need outbound forwarding. Private Resolver provides managed inbound and outbound endpoints plus rulesets. Avoid creating inconsistent duplicate zones in many subscriptions. Central DNS ownership should define zone placement, VNet links, forwarding rules, logging, and change control.

```
az network private-dns zone list -g rg-dns -o table
az network private-endpoint list -g rg-app -o table
nslookup mystorageaccount.blob.core.windows.net
```

Validation rule For a private endpoint test, confirm that the service name resolves to the assigned private IP from every required client location before disabling public access.

Hands-on Lab 16: Secure a Storage Service with Private Endpoint and Private DNS

Objective Create private access to a storage account, validate DNS and network behavior, and then disable unneeded public access.

Prerequisites

- A VNet and private endpoint subnet
- Permission to create a storage account, private endpoint, and private DNS zone

Implementation

36. Create a storage account with a unique name and a test container.
37. Create the documented private DNS zone for the Blob service and link it to the test VNet.
38. Create a Blob private endpoint and integrate it with the private DNS zone.
39. From a VM in the linked VNet, resolve the storage hostname and record the private IP.
40. Access the service using identity-based authorization rather than an account key.
41. Disable or restrict public network access after private validation.
42. Test from an unlinked or public client and document the expected failure.

Validation Evidence

- The standard service hostname resolves to the private endpoint IP inside the VNet.

- Authorized private access succeeds.
- Unauthorized public access fails after restriction.
- DNS zone link and endpoint connection state are captured.

Cleanup

- Re-enable public access only if required for safe cleanup, then remove data, endpoint, zone link, zone, storage account, VM, and resource group.
- Confirm private endpoint NICs are removed.

Stretch challenge Add a Private Resolver design for on-premises clients and document conditional-forwarding rules.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Hostname resolves to public IP inside VNet	Private zone missing, not linked, or record absent	Inspect zone, links, endpoint DNS group, and client resolver	Link correct zone and repair record integration.
DNS is private but connection fails	NSG, route, proxy, service authorization, or endpoint approval problem	Test port, endpoint state, and identity authorization	Correct network or access control without using keys.
On-premises cannot resolve private name	Conditional forwarder or resolver endpoint/ruleset missing	Trace DNS query path and forwarding rules	Implement Private Resolver or approved DNS forwarding architecture.

Interview Questions and Model Answers

1. Why is DNS critical to Private Link?

Clients normally use the service hostname; it must resolve to the private endpoint IP for traffic to take the private path.

2. What is a private endpoint?

A VNet network interface with a private IP that privately connects to a supported service through Private Link.

3. How do service endpoints differ?

They keep access on the service public endpoint but extend VNet/subnet identity; private endpoints provide a private IP in the VNet.

4. Why use alias records?

They can reference supported Azure resources and follow lifecycle changes without manually managing an IP address.

5. How do you avoid private DNS sprawl?

Centralize zone ownership, standardize links and forwarding, and automate endpoint DNS integration through policy and infrastructure as code.

Chapter Summary

- DNS changes are production changes with ownership and rollback.
- Private endpoints require correct private DNS to work reliably.
- Public access should be disabled only after every required client path is validated.

VOLUME 7B | CHAPTER 17

Chapter 17 - Azure Storage Accounts and Blob Data Engineering

Designing object storage for durability, security, lifecycle, performance, and recovery

Learning Objectives

- Explain storage account services, namespaces, account types, redundancy, and performance options.
- Design Blob containers, access tiers, lifecycle policies, versioning, soft delete, immutability, and replication.
- Use Microsoft Entra authorization, managed identity, private endpoints, and controlled SAS when delegation is unavoidable.
- Diagnose throttling, authorization, DNS, replication, lifecycle, and accidental-deletion scenarios.

17.1 Storage Account as a Security and Scale Boundary

A storage account provides namespaces and configuration for services such as Blob, Files, Queues, and Tables. Account design affects region, redundancy, performance, network exposure, encryption, identity, lifecycle, limits, and blast radius. Do not place unrelated workloads in one account solely for convenience; separate by data classification, lifecycle, performance, ownership, and recovery requirements.

Redundancy	Protection scope	Design note
LRS	Multiple copies within one datacenter.	Lowest redundancy scope; use only when accepted by workload risk.
ZRS	Synchronous copies across availability zones in the primary region.	Protects against a zone failure where supported.
GRS / RA-GRS	Primary-region replication plus asynchronous copy to paired/secondary region; RA permits secondary reads.	Regional recovery requires understanding replication lag and failover behavior.
GZRS / RA-GZRS	Zonal resilience in primary plus geo-replication.	Strong regional design where service, region, and tier support requirements are met.

17.2 Blob Lifecycle and Data Protection

Feature	Purpose	Important distinction
Hot/Cool/Cold/Archive tiers	Align storage and access cost with usage.	Archive rehydration is not immediate; minimum retention and transaction costs apply.
Lifecycle management	Moves or deletes blobs based on rules.	Test filters and scope; lifecycle deletion is not a backup strategy.
Versioning	Keeps previous blob versions after overwrite.	Can increase cost and requires retention policy.
Soft delete	Retains deleted blobs, containers, or shares for a period.	Protects against accidental deletion but must be tested.

Feature	Purpose	Important distinction
Immutable storage	WORM retention or legal hold.	Governance and compliance controls require careful authorization.
Object replication	Asynchronously copies block blobs between accounts.	Understand supported features, latency, and conflict model.

17.3 Authorization and Network Security

- Prefer Microsoft Entra authorization and Azure RBAC data roles over account keys.
- Use managed identities for workloads and user-delegation SAS when a short-lived delegated token is required.
- Disable shared-key authorization where application compatibility permits and monitor key usage during migration.
- Use private endpoints or approved network restrictions; validate DNS before disabling public access.
- Enable diagnostic settings and capture data-plane logs appropriate to security and cost requirements.

```
az storage account show -g rg-data -n <account> --query
'[{sku:sku.name,kind:kind,public:publicNetworkAccess}]'
az storage account blob-service-properties show -g rg-data -n <account>
az storage account management-policy show -g rg-data --account-name <account>
```

17.4 Performance and Cost

Performance design considers request rate, object size, concurrency, partitioning behavior, client retries, network path, and account limits. Cost includes capacity, operations, retrieval, data transfer, replication, reserved capacity, private endpoints, monitoring, and early deletion. A lifecycle policy should be based on measured access rather than age assumptions alone.

Hands-on Lab 17: Create a Secure Blob Lifecycle

Objective Build an identity-based Blob service with versioning, soft delete, lifecycle management, and private network access.

Prerequisites

- A sandbox subscription and managed-identity-enabled VM or Cloud Shell identity
- Permission to create storage, RBAC, private endpoint, and DNS resources

Implementation

43. Create a general-purpose v2 storage account with an approved redundancy option.
44. Create containers for active and archive-candidate data; keep anonymous access disabled.
45. Assign the narrow Blob data role to a managed identity or training user at the required scope.
46. Enable blob versioning and soft delete with a documented retention period.
47. Create a lifecycle policy that moves a tagged or prefix-scoped test object to a cooler tier and deletes only controlled test data.
48. Create and validate a private endpoint, then restrict public network access.
49. Upload, overwrite, delete, restore, and list versions of a test object; capture evidence and costs.

Validation Evidence

- Data operations use Entra authorization rather than account keys.
- The private hostname resolves correctly.
- Version and soft-delete recovery are demonstrated.
- Lifecycle rules target only the intended prefix or tags.

Cleanup

- Restore public connectivity only if needed for cleanup, remove objects and versions, private endpoint, DNS resources, role assignments, account, and resource group.
- Confirm no immutable retention blocks safe lab deletion.

Stretch challenge Create an event-driven process that records blob creation metadata without exposing storage keys.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
403 AuthorizationPermissionMismatch	Wrong data-plane role, scope, token, or propagation	Inspect identity, role assignment, and authentication method	Assign minimal data role at correct scope and reacquire token.
Private access works by IP but not name	Private DNS integration is wrong	Resolve service hostname and inspect private zone link	Correct zone and endpoint DNS group.
Lifecycle deletes unexpected objects	Rule filter or prefix is too broad	Review policy and object properties	Disable rule, recover through version/soft delete, and deploy tested filters.

Interview Questions and Model Answers

1. How does ZRS differ from GRS?

ZRS synchronously protects across zones in one region; GRS asynchronously replicates to a secondary region after primary-region replication.

2. Why are account keys risky?

They are broad, long-lived shared secrets that are difficult to attribute and rotate without application impact.

3. Is lifecycle management a backup?

No. It changes tier or deletes data; backup and recovery require independent retention and restore controls.

4. What protects against accidental overwrite?

Blob versioning, soft delete, tested recovery procedures, and appropriate immutable retention where required.

5. What belongs in storage cost analysis?

Capacity, operations, retrieval, early deletion, replication, transfer, monitoring, private networking, and recovery retention.

Chapter Summary

- Storage accounts are security, scale, ownership, and recovery boundaries.
- Redundancy protects different failure scopes and does not replace backup.
- Identity-based access and private networking reduce secret and exposure risk.

VOLUME 7B | CHAPTER 18

Chapter 18 - Managed Disks, Azure Files, and Enterprise File Storage

Engineering block and shared file systems for performance, availability, and recovery

Learning Objectives

- Compare managed disk types, performance properties, bursting, encryption, snapshots, and shared disks.
- Design OS and data disk layout for VM workloads and clustered applications.
- Compare Azure Files, Azure File Sync, Azure NetApp Files, and Elastic SAN scenarios.
- Troubleshoot latency, IOPS, throughput, mount, identity, quota, and snapshot failures.

18.1 Managed Disk Selection

Disk type	Typical scenario	Engineering focus
Standard HDD	Infrequent access, backup-oriented or low-performance workloads.	Latency variability and throughput limits.
Standard SSD	Web servers, development, and moderate workloads.	Balance predictable performance and cost.
Premium SSD	Production workloads requiring consistent low latency.	Size-based performance, bursting, and VM limits.
Premium SSD v2	Performance-sensitive workloads needing granular IOPS and throughput.	Regional/zonal and feature support validation.
Ultra Disk	Top-tier I/O-intensive databases and transaction systems.	Provisioned IOPS/throughput, VM compatibility, and availability design.

Application performance is limited by the lowest of disk provisioned performance, VM storage limits, controller behavior, guest filesystem, queue depth, and workload pattern. Increasing disk size does not fix inefficient synchronous I/O or VM-level caps. Measure latency, IOPS, throughput, queue depth, cache mode, and application response together.

18.2 Snapshots, Images, Backups, and Shared Disks

Mechanism	Primary use	Limitation
Managed disk snapshot	Read-only point-in-time disk copy and disk cloning.	Crash-consistency or application consistency must be planned; snapshot lifecycle is manual unless managed by backup.
VM image	Reusable OS and application baseline.	Not a recurring data backup.
Azure Disk Backup	Policy-driven snapshot lifecycle for managed disks.	Different recovery capabilities from full Azure VM Backup.
Azure VM Backup	VM-level recovery points, restore scenarios, and application-aware options.	Requires vault, policy, retention, and restore testing.

Mechanism	Primary use	Limitation
Shared disk	One managed disk attached to multiple VMs for supported clustering.	Application cluster software must coordinate writes; it is not a general shared filesystem.

18.3 Shared File Services

Service	Protocol / model	Best fit
Azure Files	Managed SMB and NFS shares.	Lift-and-shift file shares, shared application data, profiles, and cloud file services.
Azure File Sync	Caches Azure Files on Windows Servers.	Hybrid file server and distributed cache scenarios.
Azure NetApp Files	High-performance managed enterprise NFS/SMB.	Latency-sensitive enterprise file workloads and large-scale applications.
Elastic SAN	Managed SAN-like block storage for many workloads.	Consolidated block storage where supported architecture and economics fit.

18.4 Security and Operations

- Use disk encryption and customer-managed keys only with a designed key availability, rotation, and recovery process.
- Restrict disk import/export and storage access; use Private Link where supported and required.
- Use identity-based SMB access and avoid distributing storage account keys.
- Monitor disk latency, throttling, VM storage limits, share transactions, capacity, and snapshot growth.
- Test file, disk, and complete-VM recovery separately because each has different objectives.

```
az vm show -g rg-core -n vm-db-01 --query storageProfile -o json
az disk list -g rg-core -o table
az storage share-rm list -g rg-data --storage-account <account> -o table
```

Hands-on Lab 18: Attach, Benchmark, Snapshot, and Recover a Data Disk

Objective Provision a managed data disk, validate performance boundaries, create a snapshot, and recover to a new disk.

Prerequisites

- A small Linux VM
- Permission to create managed disks and snapshots
- A safe benchmark tool and explicit performance limit

Implementation

50. Create a managed data disk appropriate for the lab and attach it to the VM.
51. Identify the device safely, partition and format it, mount by UUID, and persist the mount configuration.
52. Run a bounded benchmark and record IOPS, throughput, latency, VM size, disk SKU, and cache mode.

53. Write controlled test data and quiesce the application or filesystem as appropriate.
54. Create a snapshot and use it to create a replacement disk.
55. Attach and mount the recovered disk to a recovery VM or alternate mount point.
56. Validate file integrity and document RPO/RTO observations and cleanup.

Validation Evidence

- The mount survives a VM restart.
- Benchmark results are interpreted against both disk and VM limits.
- Recovered data matches the expected point in time.
- No credentials or broad public storage access are used.

Cleanup

- Unmount safely, detach and delete recovered and source lab disks, snapshots, VM, and resource group.
- Confirm no orphaned disks remain.

Stretch challenge Create an Azure Files share using identity-based access and compare latency and semantics with the managed disk.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Disk performance below expectation	VM cap, disk cap, queue depth, cache, filesystem, or workload pattern	Compare metrics at disk, VM, guest, and application layers	Resize or tune the correct bottleneck and retest.
Disk appears with unexpected device name	Linux device enumeration changed	Use lsblk, UUID, and filesystem metadata	Mount by UUID and avoid assumptions such as /dev/sdc.
SMB mount denied	Identity, domain integration, RBAC/share ACL, DNS, or port path incorrect	Inspect authentication and network path	Correct identity/ACL and private DNS/firewall configuration.

Interview Questions and Model Answers

1. What limits managed-disk performance?

The effective limit is constrained by disk provisioning, VM storage limits, controller, guest configuration, and application I/O pattern.

2. How is a snapshot different from a backup?

A snapshot is a point-in-time disk copy; a backup adds policy, retention, protection, orchestration, and broader restore workflows.

3. When use Azure Files instead of a data disk?

When multiple clients need managed SMB/NFS file semantics rather than VM-attached block storage.

4. What is a shared disk?

A managed block disk attachable to multiple VMs for supported clustered applications; cluster software coordinates access.

5. Why mount Linux data disks by UUID?

Device names can change across boots or VM configurations, while filesystem UUID provides stable identification.

Chapter Summary

- Disk choice must consider both disk and VM performance ceilings.
- Snapshots, images, disk backup, and VM backup have different recovery purposes.
- Azure Files and Azure NetApp Files provide shared file semantics; managed disks provide block storage.
- Recovery must be validated through an actual restore.

VOLUME 7B | CHAPTER 19

Chapter 19 - Hybrid Connectivity with VPN Gateway, ExpressRoute, and Virtual WAN

Connecting branches, datacenters, users, and Azure networks with resilient routing

Learning Objectives

- Compare site-to-site VPN, point-to-site VPN, ExpressRoute, VNet-to-VNet connectivity, and Virtual WAN.
- Design gateway subnets, SKUs, active-active modes, BGP, route propagation, and coexistence.
- Plan hybrid DNS, firewall inspection, redundancy, monitoring, and failure testing.
- Diagnose tunnel, BGP, route, MTU, asymmetric path, throughput, and name-resolution issues.

19.1 Connectivity Option Matrix

Option	Transport	Best fit
Site-to-site VPN	Encrypted IPsec/IKE over internet	Branches, datacenters, backup path, and moderate hybrid connectivity.
Point-to-site VPN	Encrypted user/device tunnel	Administrative and remote-user access to private Azure resources.
ExpressRoute	Private provider connectivity to Microsoft network	Predictable private enterprise connectivity and high-throughput hybrid scenarios.
VNet peering	Microsoft backbone between VNets	Low-latency Azure VNet connectivity within supported topology.
Virtual WAN	Microsoft-managed hubs integrating branches, users, ExpressRoute, VPN, and VNets	Large-scale global transit and managed connectivity operations.

19.2 Gateway and Route Design

GatewaySubnet size, SKU, zone redundancy, active-active configuration, BGP, route limits, and throughput must be selected before workload onboarding. BGP automates route exchange but does not replace route governance. Advertised prefixes, local network gateway definitions, autonomous system numbers, route filters, and propagation must be reviewed as code or controlled configuration.

ExpressRoute and VPN can coexist for migration or backup, but path preference and failover behavior must be understood. Asymmetric paths through stateful firewalls cause silent failures. A hybrid design must define the primary path, secondary path, convergence target, expected packet loss, and operational authority during provider or on-premises incidents.

19.3 Hybrid DNS and Identity Dependencies

- Provide conditional forwarding for Azure private zones and on-premises namespaces.
- Use Private Resolver or an approved DNS architecture with resilient endpoints and monitoring.
- Ensure domain controllers, certificate authorities, identity services, and time sources have redundant paths.

- Prevent accidental route propagation from exposing management or restricted networks.
- Log gateway health, BGP peers, connection state, bandwidth, packet drops, DNS queries, and firewall decisions.

```
az network vpn-connection list -g rg-connectivity -o table
az network vnet-gateway list-bgp-peer-status -g rg-connectivity -n vng-hub -o table
az network express-route list -g rg-connectivity -o table
```

19.4 Failure Testing

A green tunnel status is not proof of application continuity. Test traffic from representative subnets, large packets, long-lived connections, DNS, authentication, and application dependencies. During a controlled failover, record detection time, routing convergence, packet loss, reconnection behavior, and return to the preferred path.

Hands-on Lab 19: Design and Validate Hybrid Connectivity

Objective Create a cost-controlled VPN lab or a complete design simulation, validate routes and failover assumptions, and produce an operational runbook.

Prerequisites

- An on-premises lab VPN device, simulated network, or instructor-provided endpoint
- Permission and budget for VPN Gateway if deployed
- Nonoverlapping Azure and on-premises CIDRs

Implementation

57. Document business requirements, traffic flows, availability, throughput, encryption, and recovery objectives.
58. Create a hub VNet with a correctly sized GatewaySubnet.
59. Deploy an approved VPN Gateway SKU or produce a command-complete simulation when cost prevents deployment.
60. Configure local network gateway, connection, shared secret handling, and BGP if used.
61. Validate advertised and effective routes from representative workloads.
62. Test DNS and application connectivity in both directions.
63. Create a failover and troubleshooting runbook including tunnel, route, MTU, DNS, and firewall checks.

Validation Evidence

- Address ranges do not overlap.
- The route table contains only intended hybrid prefixes.
- Secrets are handled through an approved method and are not stored in the report.
- The runbook identifies owners, evidence, escalation, and rollback.

Cleanup

- Delete VPN connections, gateways, public IPs, local network gateways, test VNets, and resource groups if deployed.
- Confirm gateway resources have completed deletion because they can continue to incur charges.

Stretch challenge Design ExpressRoute plus VPN backup and explain route preference and failure behavior.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Tunnel is down	IKE/IPsec mismatch, public IP, shared secret, or device reachability	Compare both endpoint configurations and gateway diagnostics	Align proposals and verify public path and credentials.
Tunnel is up but subnet unreachable	Missing/incorrect prefixes, UDR, NSG, BGP, or return path	Inspect effective routes and advertised/learned prefixes	Correct route advertisement and symmetric return path.
Small packets work, application fails	MTU/MSS, fragmentation, firewall timeout, or DNS issue	Test packet sizes, captures, and application logs	Tune supported MTU/MSS and correct stateful path or DNS.

Interview Questions and Model Answers

1. When is ExpressRoute preferable to VPN?

When private provider connectivity, predictable enterprise routing, throughput, or compliance requirements justify the service and dual-path design.

2. What does BGP provide?

Dynamic prefix exchange and route selection, reducing manual route maintenance while requiring strict advertisement governance.

3. Why does a tunnel-up state not prove success?

It proves control-plane establishment, not application routes, DNS, security, MTU, return path, or dependency health.

4. What causes asymmetric-routing failures?

Forward and return traffic traverse different stateful gateways or firewalls, so connection state is missing on one path.

5. What must a hybrid failover test record?

Detection, route convergence, packet loss, connection recovery, DNS and identity behavior, application impact, and restoration to primary.

Chapter Summary

- Hybrid connectivity selection depends on transport, scale, performance, and operations.
- BGP automates exchange but route ownership and filtering remain essential.
- DNS and identity are critical hybrid dependencies.
- Controlled failover tests must validate applications, not only tunnel state.

VOLUME 7B | CHAPTER 20

Chapter 20 - Azure Backup, Site Recovery, and Core Infrastructure Capstone

Turning recovery objectives into tested protection and an integrated production design

Learning Objectives

- Distinguish high availability, backup, replication, disaster recovery, and business continuity.
- Design vaults, policies, retention, immutability, soft delete, encryption, and access separation.
- Use Azure Site Recovery concepts, recovery plans, test failover, failover, and reprotection.
- Integrate compute, network, storage, DNS, application delivery, hybrid connectivity, monitoring, and recovery into one capstone.

20.1 Recovery Controls Solve Different Failures

Control	Protects against	Does not automatically protect against
High availability	Instance, zone, or component failure within the designed scope.	Data corruption, deletion, ransomware, or regional disaster unless separately designed.
Backup	Deletion, corruption, ransomware, and point-in-time recovery within retention.	Immediate service continuity without restore and application recovery steps.
Replication / Site Recovery	Workload continuity in an alternate site or region.	Bad changes or corruption that replicate to the target without independent backup.
Immutable retention	Deletion or alteration of protected recovery data during retention.	Application consistency and tested restoration by itself.
Business continuity plan	People, process, technology, communications, and priorities.	Technical recovery unless the infrastructure and procedures are implemented and exercised.

20.2 Azure Backup Architecture

Azure Backup uses vaults and workload-specific policies to create and retain recovery points. Design decisions include vault type and region, redundancy, cross-region restore requirements, enhanced security, soft delete, immutability, encryption, private access, identity, policy frequency, retention, and operational separation. Backup operators should not automatically have rights to alter production resources or delete protected data.

20.3 Site Recovery and Recovery Plans

Azure Site Recovery orchestrates replication and failover for supported workloads. The recovery design defines source and target regions, replication settings, target networks, resource groups, capacity, IP/DNS behavior, dependency order, automation, test network, failover, commit, reprotection, and fallback. A recovery plan groups machines and actions in ordered stages, but application teams still own data and functional validation.

20.4 RPO, RTO, and Recovery Evidence

Term	Question	Evidence
RPO	How much recent data loss can the business tolerate?	Recovery-point timestamps, replication lag, backup frequency, and application transaction checks.
RTO	How long can the service be unavailable?	Measured detection, decision, restore/failover, DNS, startup, validation, and communication time.
Restore success	Can protected data be recovered?	Completed restore, integrity validation, application start, and user transaction.
DR success	Can the service operate in the recovery location?	Isolated test failover, dependency validation, performance, security, and runbook timing.

```
az backup vault list -o table
az backup job list --resource-group rg-recovery --vault-name rsv-core -o table
az site-recovery job list -g rg-recovery --vault-name rsv-dr -o table
```

20.5 Capstone Architecture

Scenario Design a secure customer-facing application for a regulated company. It requires zone-resilient compute, private data access, controlled internet entry, centralized inspection, hybrid administration, defined backup, and tested regional recovery. The workload must be reproducible through infrastructure as code in a later volume.

Layer	Required design outcome
Identity and governance	Managed identities, RBAC groups, policy guardrails, tags, locks, budgets, activity logs.
Compute	Scale set across zones, health-aware autoscale, versioned image, rolling upgrade, secure administration.
Network	Hub-spoke, nonoverlapping CIDRs, NSGs/ASGs, inspected egress, private DNS, Bastion.
Application delivery	Front Door or appropriate global entry, regional Application Gateway/Load Balancer, WAF, TLS, protected origins.
Storage	Private storage, identity authorization, versioning/soft delete, lifecycle, backup separate from replication.
Hybrid	VPN/ExpressRoute design, BGP and route governance, hybrid DNS, backup path, monitoring.
Recovery	Documented RPO/RTO, vault security, restore test, isolated Site Recovery test failover, runbooks and evidence.

Hands-on Lab 20: Core Infrastructure Capstone and Recovery Exercise

Objective Produce and validate an enterprise Azure infrastructure design that integrates Chapters 11-20 and proves one restore and one controlled failover scenario.

Prerequisites

- Completion of Part B labs or equivalent knowledge
- A sandbox environment or architecture-simulation approval
- A defined workload scenario and business objectives

Implementation

64. Write requirements, assumptions, constraints, data classification, RPO, RTO, regions, quotas, and cost limits.
65. Create the logical architecture, subscription/resource-group placement, address plan, hub-spoke routing, security flows, DNS, and ownership matrix.
66. Deploy a cost-controlled implementation with two or more healthy application instances or a scale set behind the chosen delivery service.
67. Use managed identity and private endpoint access for storage; enable versioning/soft delete and a tested lifecycle rule.
68. Configure backup for a selected workload and perform a restore into an isolated validation location.
69. Configure Site Recovery for a supported lab VM or produce an exact recovery-plan simulation; perform test failover where feasible.
70. Run failure scenarios: unhealthy backend, denied network flow, private DNS fault, lost data object, and primary workload outage.
71. Capture metrics, logs, screenshots, commands, timings, findings, and cost; remediate defects and repeat failed tests.
72. Produce an operations handover with dashboards, alerts, patching, backup, DR, incident, ownership, and cleanup procedures.

Validation Evidence

- Architecture traces every requirement to a control and an owner.
- Application traffic survives a backend failure.
- Private storage access and public restriction are proven.
- A real restore is validated for integrity and application use.
- Recovery timing is measured against RTO and data point against RPO.
- All lab resources are inventoried and cost ownership is visible.

Cleanup

- Execute the teardown runbook in dependency order.
- Confirm gateways, firewalls, Front Door/Application Gateway, public IPs, private endpoints, disks, snapshots, backups, replication, vaults, and DNS records are removed or intentionally retained.
- Export final cost and deletion evidence.

Stretch challenge Rebuild the entire capstone from Bicep or Terraform and use an automated validation suite to test routes, private DNS, health, recovery points, and policy compliance.

20.6 Recovery Runbook Template

Phase	Actions	Exit evidence
Declare	Confirm incident, authority, scope, business priority, and communication channel.	Incident record and decision owner.
Assess	Validate source, backup/replication state, dependency health, target capacity, and security.	Recovery option and risk statement.
Recover	Restore or test/fail over in dependency order; update routing/DNS only under approval.	Infrastructure and application health.
Validate	Run data-integrity, identity, security, performance, and business-transaction tests.	Business owner acceptance.
Stabilize	Monitor, scale, protect backups, communicate, and plan reprotection/failback.	Stable service and documented next action.
Learn	Reconcile data, close gaps, update architecture/runbook, and track actions.	Post-incident review with owners and dates.

Troubleshooting Matrix

Symptom	Likely cause	Investigation	Corrective action
Backup succeeds but restore fails validation	Recovery point, application consistency, dependencies, or encryption access problem	Inspect job, restored data, logs, keys, and dependency versions	Select correct recovery point and restore required dependency chain.
Test failover VM starts but application is down	Target network, DNS, identity, load balancer, secrets, or capacity missing	Validate recovery plan and target dependency map	Provision and test complete recovery environment, not only VM replication.
Vault cannot be deleted	Protected items, soft-deleted items, immutability, private endpoints, or registered servers remain	Inventory vault dependencies and protection state	Follow safe disable-protection and retention procedure; never bypass compliance controls.

Interview Questions and Model Answers

1. How is backup different from Site Recovery?

Backup retains recovery points for restore; Site Recovery replicates and orchestrates workload continuity in an alternate location. Both may be required.

2. Why is replication not a backup?

Corruption or destructive changes can replicate to the target, while backup preserves independent historical recovery points.

3. What is test failover?

An isolated recovery exercise that validates replicated workloads without disrupting production replication or initiating planned/unplanned failover.

4. How do you prove an RTO?

Measure the full timeline from detection and decision through recovery, DNS/routing, application validation, and business acceptance.

5. What should be separated in backup administration?

Production operations, backup policy management, restore execution, deletion authority, key management, and audit should follow least privilege and separation of duties.

Chapter Summary

- Availability, backup, replication, and continuity address different failure modes.
- Recovery objectives must be translated into retention, topology, capacity, runbooks, and tests.
- A completed backup job is not evidence of recoverability until restore validation succeeds.
- Part B is complete when an integrated infrastructure design survives controlled failures and produces auditable evidence.

VOLUME 7B | CHAPTER B1

Chapter B1 - Part B Command Reference

High-value discovery and troubleshooting commands

Compute and Scale Sets

```
az vm list -d -o table
az vm get-instance-view -g <rg> -n <vm> -o json
az vmss list -g <rg> -o table
az vmss list-instances -g <rg> -n <vmss> -o table
az monitor autoscale list -g <rg> -o table
```

Networking and Security

```
az network vnet list -o table
az network vnet peering list -g <rg> --vnet-name <vnet> -o table
az network nic show-effective-route-table -g <rg> -n <nic> -o table
az network nic list-effective-nsg -g <rg> -n <nic> -o json
az network watcher test-ip-flow ...
az network watcher show-next-hop ...
```

Application Delivery and DNS

```
az network lb list -o table
az network application-gateway show-backend-health -g <rg> -n <agw>
az afd profile list -o table
az network dns zone list -o table
az network private-dns zone list -o table
az network private-endpoint list -o table
```

Storage and Recovery

```
az storage account list -o table
az disk list -o table
az snapshot list -o table
az backup vault list -o table
az backup job list -g <rg> --vault-name <vault> -o table
az site-recovery job list -g <rg> --vault-name <vault> -o table
```

VOLUME 7B | CHAPTER B2

Chapter B2 - Production Readiness Checklist

Evidence required before an Azure core-infrastructure workload is approved

Domain	Readiness evidence
Requirements	Workload owners, data classification, SLO, RPO, RTO, regions, zones, quotas, and cost limits approved.
Compute	Versioned images, health checks, autoscale, rolling upgrade, patching, secure administration, and replacement test.
Network	Address plan, hub-spoke topology, route ownership, effective route/rule evidence, private administration, and hybrid paths.
Security	Managed identity, least privilege, NSG/ASG rules, inspected egress, WAF where applicable, and logging.
Delivery	Correct service selection, health probes, TLS, origin protection, backend drain, and failure test.
DNS/private access	Zone ownership, private endpoint DNS, hybrid forwarding, public restriction, and negative-path tests.
Storage	Redundancy, identity access, encryption, versioning/soft delete, lifecycle, capacity, and cost controls.
Recovery	Independent backup, immutable/soft-delete controls, successful restore, DR test, measured RPO/RTO, and runbooks.
Operations	Dashboards, alerts, logs, patching, incident process, owner contacts, change process, and cleanup/retirement procedure.

VOLUME 7B | CHAPTER B3

Chapter B3 - Glossary

Core infrastructure terms

Term	Definition
Application Gateway	Regional Layer 7 web traffic load balancer with routing and optional WAF.
Application security group	Logical grouping of NICs used as source or destination in NSG rules.
Azure Bastion	Managed service for private RDP/SSH administration through the Azure portal or supported clients.
Azure Firewall	Managed stateful network security service for centralized network and application traffic control.
Azure Front Door	Global HTTP(S) edge, acceleration, routing, failover, caching, and WAF service.
BGP	Dynamic routing protocol used to exchange prefixes between network systems.
Hub-spoke	Topology with shared connectivity/security in hubs and isolated workload VNets in spokes.
Managed disk	Azure-managed block storage for virtual machines.
NSG	Stateful Layer 3/4 network filter associated with subnets or NICs.
Private endpoint	Private IP interface in a VNet connecting to a service through Private Link.
Private Resolver	Managed Azure DNS forwarding service with inbound and outbound endpoints.
RPO	Maximum tolerable data loss measured in time.
RTO	Maximum tolerable service restoration time.
Site Recovery	Azure disaster-recovery replication and failover orchestration service.
Traffic Manager	DNS-based global traffic distribution service.
UDR	User-defined route that overrides selected Azure system-routing behavior.
Virtual Machine Scale Set	Azure orchestration resource for creating, distributing, scaling, and upgrading VM fleets.
Virtual WAN	Microsoft-managed networking service for global hub, branch, user, ExpressRoute, VPN, and VNet connectivity.

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Chapter B4 - Official Microsoft Documentation Index

Primary sources for continued study

Azure Virtual Machines: <https://learn.microsoft.com/azure/virtual-machines/>

Virtual Machine Scale Sets: <https://learn.microsoft.com/azure/virtual-machine-scale-sets/>

Azure networking design guide: <https://learn.microsoft.com/azure/networking/design-guide/>

Hub-spoke architecture: <https://learn.microsoft.com/azure/architecture/networking/architecture/hub-spoke>

Network security groups: <https://learn.microsoft.com/azure/virtual-network/network-security-groups-overview>

Azure Firewall: <https://learn.microsoft.com/azure/firewall/overview>

Load balancing options: <https://learn.microsoft.com/azure/architecture/guide/technology-choices/load-balancing-overview>

Azure DNS: <https://learn.microsoft.com/azure/dns/>

Azure Private Link: <https://learn.microsoft.com/azure/private-link/>

Azure Storage: <https://learn.microsoft.com/azure/storage/>

Managed disks: <https://learn.microsoft.com/azure/virtual-machines/managed-disks-overview>

Azure Files: <https://learn.microsoft.com/azure/storage/files/>

VPN Gateway: <https://learn.microsoft.com/azure/vpn-gateway/>

ExpressRoute: <https://learn.microsoft.com/azure/expressroute/>

Virtual WAN: <https://learn.microsoft.com/azure/virtual-wan/>

Azure Backup: <https://learn.microsoft.com/azure/backup/>

Azure Site Recovery: <https://learn.microsoft.com/azure/site-recovery/>